

Spring 2026. Physics 510.
Quantum Theory II.

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Homework # 6. Due to April 29th, 2026.

Dirac gamma-matrices.

Dirac or Standard representation:

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \gamma^5 = -\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}.$$

Here I is the unit 2-dimensional matrix and σ^i are Pauli matrices. Quite frequently γ^i is defined with the opposite sign $\gamma^i = \begin{pmatrix} 0 & -\sigma^i \\ \sigma^i & 0 \end{pmatrix}$, and therefore γ^5 should be defined as $\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$.

Weyl or Chiral representation:

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \gamma^i = \begin{pmatrix} 0 & -\sigma^i \\ \sigma^i & 0 \end{pmatrix}, \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}.$$

Majorana representation:

$$\gamma^0 = \begin{pmatrix} 0 & \sigma^2 \\ \sigma^2 & 0 \end{pmatrix}, \gamma^1 = \begin{pmatrix} i\sigma^3 & 0 \\ 0 & i\sigma^3 \end{pmatrix}, \gamma^2 = \begin{pmatrix} 0 & -\sigma^2 \\ \sigma^2 & 0 \end{pmatrix}, \gamma^3 = \begin{pmatrix} -i\sigma^1 & 0 \\ 0 & -i\sigma^1 \end{pmatrix}$$

All components of γ -matrices, including γ^5 , are pure imaginary in Majorana representation.

1. Find eigenvalues and eigenvectors of γ^0 and the matrix of transformation from the Weyl representation to the Dirac one that diagonalizes γ^0 . Extra bonus: $U = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ I & -I \end{pmatrix}$ is a wise choice. Then, calculate γ -matrices in the Dirac representation using the matrix of transformation.
2. Find eigenvalues and eigenvectors of γ^1 and the matrix of transformation from the Dirac representation to the Majorana one that diagonalizes γ^1 . Extra bonus: $U = \frac{1}{\sqrt{2}} \begin{pmatrix} I & \sigma^2 \\ \sigma^2 & -I \end{pmatrix}$ is a wise choice. Then, calculate γ -matrices in the Majorana representation using the matrix of transformation.
3. Calculate $Tr(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\alpha\gamma^\beta)$.
4. Calculate $\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu$ and $\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\nu\gamma_\mu$.
5. Calculate the trace $Tr\{\gamma^\mu(\hat{p} + m)(\hat{q} + m)\gamma^\nu\}$, where $\hat{p} = p^\alpha\gamma_\alpha$.
6. Introduce $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu]$. How many components does it have in the Minkowski space-time?
7. Show that $\sigma^{\mu\nu}\sigma_{\mu\nu} = 12$. Calculate the commutator $[\sigma^{\mu\nu}, \sigma^{\alpha\beta}]$.